HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **C++** modules for a graphics profiling tool, focusing on **performance analysis** and data collection from GPU instrumentation.
- Developed **Python** scripts for automating data ingestion and pre-processing of graphics performance logs, ensuring data integrity for downstream analysis.
- Collaborated with GPU hardware engineers to integrate low-level **driver instrumentation** for real-time performance metrics collection.
- Engineered features for a **graphics debugging tool**, enabling developers to step through execution and inspect shader states.
- Optimized the user experience of the tool's **profiling interface**, reducing data loading times by **30%** through efficient data handling.
- Participated in code reviews and technical discussions to ensure code quality and adherence to **C++** best practices for tool development.
- Containerized the graphics tool suite using **Docker**, streamlining deployment and ensuring consistent execution across development and testing environments.

SKILLS

Languages – C++, Python, C#, Rust, Go

Graphics APIs – Vulkan, DirectX, Metal, OpenGL, WebGL

GPU Computing – OpenCL, CUDA, SYCL

Tools & Debugging – Nsight, PIX, GPA, RenderDoc, Debuggers, Profilers

Graphics Concepts – Ray Tracing, Mesh Shading, Work Graphs, Image Processing, Data Parallelism, Machine Learning Algorithms

Development Platforms – Unity, Unreal Engine

Core Technologies – Linux, Docker, Git, CI/CD

PROJECTS

GPU Performance Analysis Tool

- Developed a **C++** application to capture and analyze GPU performance metrics, integrating with custom driver hooks.
- Implemented data visualization components to present performance bottlenecks and suggest optimization strategies to developers.
- Designed the tool's architecture to support future extensions for **ray tracing** and **mesh shading** analysis.

Graphics Debugger Extension

- Built a plugin for an existing **graphics debugger** to provide enhanced visualization of **real-time graphics** pipeline states.
- Wrote **Python** scripts to automate the generation of test cases for validating debugger accuracy against known rendering issues.
- Collaborated with a small team to ensure seamless integration and end-user usability for the extension.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)